Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/16/2008
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Project Title:	Nutritional Countermeasures to Radiation-Enhanced Colon Cancer		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	77843-2253	**Congressional District:**	17
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Braby, Leslie (Texas Engineering Experiment Station) Chapkin, Robert (Texas A&M University) Ford, John (Texas A&M University) Turner, Nancy (Texas A&M University) Carroll, Raymond (Texas A&M University)		
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	1. Original Aims. The primary goal of this project is to develop a set of colon cell gene expression profiles that accurately characterize the radiation-enhancing effect on colon tumor development and the chemoprotective effect of dietary countermeasures in this process. This goal is addressed by specific aim #1: From the global transcriptional profile (analyzed by microarray technology) in samples isolated from colonic mucosa, determine which sets of gene expression profiles best characterize the initiation stage, the promotion stage (aberrant crypt development) and the final tumor stage of colon cancer development using a 2 x 4 x 4 experimental design (+ or – radiation; four time points and 4 diets). The secondary goal, addressed by specific aim #2, is to compare gene expression profiles developed from exfoliated cells to those from colonic mucosal cells to determine how well they reflect in vivo events. This information is important for our short-term future goal of using fecal material from humans as a noninvasive method to monitor changes in gene expression patterns over time. Our long-term future goal is to have sets of gene expression profiles that can be used to monitor astronauts before, during, and after space flight to detect exposure and response to a radiation event so that appropriate intervention strategies (e.g., dietary countermeasures) can be implemented as needed. Data generated from this research program will support the NASA/NSBRI goals of: 1) gaining greater understanding of the mechanisms involved in the synergy of radiation and other environmental insults on carcinogenesis, 2) developing a system to monitor the need for countermeasure administration during or after flight, and 3) development of countermeasures that can be applied before, during and after flight. 2. Key Findings. Analysis of fecal samples collected over time indicated there was a variety of genes with differences in expression in response to diet, radiation treatment, stage of disease progression and their interactions. Radiation exposure caused changes in expression for 125 genes; time of sampling affected the expression of 369 genes; and diet caused differential expression in 387 genes. Of the differentially expressed genes with a known function in the two radiation exposure groups, there were diverse functional activities represented. Expression of genes involved in metabolism, regulation of cell proliferation and differentiation, signaling cascades in colonocytes were affected by radiation exposure. After pathway analyses, many genes were identified that were over-represented for their respective biological processes. There were more than 300 genes for which differences in gene expression were observed for the diet*time interaction, indicating the transformations occurring in the colon during tumor development were sensitive to dietary intervention. These proteins are important in buffering cellular redox status, immune function, regulation of nutrient and xenobiotic metabolism, and signaling through protein kinase pathways.
Task Description:	Gene expression in mucosal samples collected at termination for the initiation, promotion and tumor stages identified treatment-induced differential expression patterns that were reflective of the particular endpoint being measured and treatment groups. For example, 10 days after radiation exposure genes known to be affected by radiation in cell culture studies that are involved in the regulation of cell cycle activity were differentially expressed. Similarly, we have previously demonstrated a reduction in aberrant crypt foci of rats consuming pectin diets and that this effect was paralleled by an increase in apoptosis. In the mucosal gene expression patterns, we have noted large numbers of genes differentially expressed in rats consuming the two fiber types, several of which are involved in apoptosis regulation. We are in the final stages of establishing gene sets that best reflect/describe the animal phenotypes. These genes can then be used to diagnose radiation exposure, monitor disease development and to target appropriate dietary interventions. Because these genes are the subjects of a patent application, we are not able to disclose their identities. However, upon submission of the patent application this information will be published. In 2008, results of this experiment were presented at the: 1) NASA Human Research Program Investigator's Workshop, February 2008; 2) Experimental Biology, April 2008; 3) 79th Annual meeting of the Aerospace Medical Association, May 11-15, Boston, MA; and 4) 37th COSPAR Scientific Assembly, July 2008, Montreal, Canada. 3. Impact of Findings on Project Goals. The observations support the feasibility of using fecal polyA+ RNA as a non-invasive measure of colonocyte gene expression in Earth-based medicine and to monitor in-flight radiation exposure and cell responses. It would be possible to identify interventions for astronauts in order to slow disease progression until their return to Earth. This work also indicates dietary modifications influence gene expression in colonocytes and the pattern of changes in expression that occur during disease progression. It is therefore possible to use diet as a mediator in colon carcinogenesis in order to reduce the enhancement of cancer risk posed by exposure to galactic cosmic radiation. We have identified genes that are most indicative of radiation exposure and of each stage of disease progression, which was the overall goal of the proposed work. 4. Proposed Research Plan for the Coming Year. We will complete the final statistical analyses and prepare the list of genes that are most predictive of radiation exposure and disease progression. The data will be incorporated into publications, which will be submitted once the patent application has been submitted.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary deterrent to great reductions in the number of deaths from colon cancer is getting individuals to undergo screening. Data generated by this project is being extended to Earth-based medical care by providing a system of detection that is non-invasive, which will facilitate patient compliance with screening recommendations. Global gene expression profiles have been determined for individuals over time in order to monitor colon cancer development, and to determine the changes in expression generated by radiation exposure and dietary intervention. Genes have identified that indicative of the stages of colon carcinogenesis and to radiation exposure, making it possible to monitor disease onset and progression using a noninvasive procedure.
Task Progress:	The primary aim of this project was to understand the changes in gene expression that occur in the colon after exposure to radiation, followed by injection of a chemical carcinogen (the chemical carcinogen is a model of sporadic colon cancer in humans). One goal is to understand disease progression and to identify gene expression patterns that are indicative of the disease stage. Our second goal is to compare gene expression profiles of mucosal samples to samples of RNA from sloughed cells in order to develop a non-invasive procedure of detecting the changes. Lastly, we wanted to understand how diets containing either corn oil or fish oil as the lipid component and either cellulose or pectin as the fiber source influence the response to radiation and carcinogen exposure, and thus disease progression. All samples have been analyzed and the data evaluated using standard statistical techniques. We are currently in the final stages of identifying genes that are most indicative of disease phenotype, preparing manuscripts and a patent application.
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